

Question No. 1 #665df48dca324ed4a4c50fbc1

Section Details

Marking Scheme >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A body starts from rest and moves with uniform acceleration. If the distance travelled by it in the first 2 s is x_1 and in the next 2 s is x_2 , then x_1 and x_2 are related as

- ☐ $x_1 = x_2$
- ☐ $x_1 = 2x_2$
- ☐ $2x_1 = x_2$
- ☒ $3x_1 = x_2$

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PHYSICS SINGLE CORRECT

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Question No. 2 #665df48dca324ed4a4c50fbe2

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Statement (A): When the temperature increases the viscosity of gases increases and the viscosity of liquids decreases.

Statement (B): Water does not wet an oily glass because cohesive force of oil is less than that of water.

Statement (C) : A liquid will wet a surface of a solid if the angle of contact is greater than 90° .

- ☐ A, B, and C are false
- ☐ A and B false, C is true
- ☒ B and C false, A is true
- ☐ A and C false, B is true

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Answered and Marked for Review (will
be considered for evaluation)**PHYSICS SINGLE CORRECT**

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Question No. 3 #665df48dca324ed4a4c50fbe3

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The refractive index of a prism for a monochromatic wave is $\sqrt{2}$ and its refracting angle is 60° . For minimum deviation, the angle of incidence will be

- ☐ 30°
- ☐ 45°
- ☒ 60°
- ☐ 75°

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0 Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Statement (I) : A uniform electric field and a uniform magnetic field are pointed in the same direction. If an electron is projected in the same direction the electron velocity will decrease in magnitude.

Statement (II) : Two infinite long parallel wires are carrying equal current in the same direction. The magnetic field at a point midway between the wires is zero.

Statement (III) : No net force acts on a rectangular coil carrying a steady current when suspended in a uniform magnetic field.

Which of the following is correct?

- ☒ Statements I, II and III are true.
- ☐ Statements I and II are true, but statement III is false.
- ☐ Statements II and III are true, but statement I is false.
- ☐ Statements I and III are true, but statement II is false.

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PHYSICS SINGLE CORRECT

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Question No. 5 #665df48dca324ed4a4c50fbc5

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The escape velocity for the earth is V_{es} . Find out escape velocity for a planet whose radius is four times and density is nine times that of the earth. If is $n \times 6 V_{es}$. Find n .

- ☐ 6
- ☐ 4
- ☐ 8
- ☒ 2

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Answered and Marked for Review (will
be considered for evaluation)**PHYSICS SINGLE CORRECT**

Choose a question

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Question No. 6 #665df48dca324ed4a4c50fbc6

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A railway track is banked for a speed v , by making the height of the outer rail h higher than that of the inner rail. If the distance between the rails is l and the radius of curvature of the track is r , then

- ☐ $\frac{h}{l} = \frac{v^2}{rg}$
- ☒ $\tan \left\{ \sin^{-1} \left(\frac{h}{l} \right) \right\} = \frac{v^2}{rg}$
- ☐ $\tan^{-1} \left(\frac{h}{l} \right) = \frac{v^2}{rg}$
- ☐ $\frac{h}{r} = \frac{v^2}{lg}$

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Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

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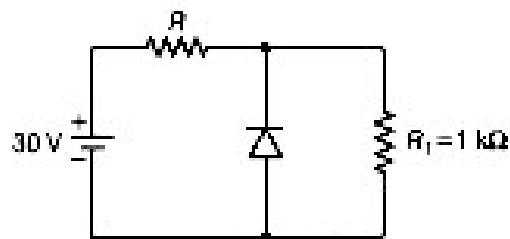
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Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If current in diode is five times that in R_1 and breakdown voltage of diode is 6 V , find the value of R ?



- ☐ 2000Ω
- ☒ $\frac{2000}{3}\Omega$
- ☐ 1000Ω
- ☐ $\frac{1000}{3}\Omega$

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Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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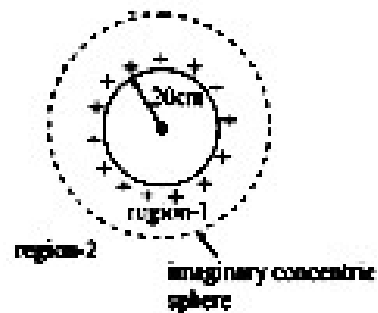
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Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

What is the radius of the imaginary concentric sphere that divides the electrostatic field of a metal sphere of a radius 20 cm and a charge of $8\mu\text{C}$ in two regions of identical energy. If it is $n \times 10$ cm then find n .



- ☐ 1
- ☐ 2
- ☐ 3
- ☒ 4

Mark for Review & Next

Clear Response

PHYSICS SINGLE CORRECT

Choose a question

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Question No. 9 #665df48dca324ed4a4c50fbe9

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements is wrong?

- ☐ KE of a body is independent of the direction of motion
- ☐ In an elastic collision of two bodies the momentum and energy of system is conserved
- ☐ If two protons are brought towards each other the PE of the system increases.
- ☒ A body cannot have energy without momentum.

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Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

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Question No. 10 #665df48dca324ed4a4c50fbe10

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A gas in container A is in thermal equilibrium with another gas in container B. Both contain equal masses of the two gases in the respective containers. Which of the following can be true?

- ☐ $P_A V_A = P_B V_B$
- ☒ $P_A = P_B, V_A \neq V_B$
- ☐ $P_A = P_B, V_A = V_B$
- ☐ $\frac{P_A}{V_A} = \frac{P_B}{V_B}$

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Answered and Marked for Review (will
be considered for evaluation)**PHYSICS SINGLE CORRECT**

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Question No. 11 #665df48dca324ed4a4c50fbc11

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let A_n be the area enclosed by the n^{th} orbit in a hydrogen atom. The graph of $\ln(A_n/A_1)$ against $\ln(n)$

- ☐ will not pass through the origin
- ☒ will have certain points lying on a straight line with slope 4
- ☐ will be a monotonically increasing non linear curve
- ☐ will be a circle

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9 Not Visited 0 Marked for Review

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PHYSICS SINGLE CORRECT

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Only One Option Correct Type



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PHYSICS SINGLE CORRECT

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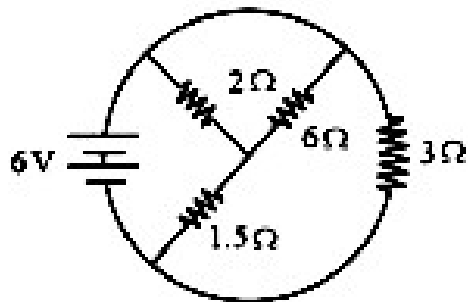
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The total current supplied to the circuit by the battery is

☐ 1 A☐ 2 A☒ 4 A☐ 6 A

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Question No. 13 #665df48dca324ed4a4c50fbc13

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

On six complete rotations, the screw gauge moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is

- ☐ 0.01 cm
- ☐ 0.02 mm
- ☐ 0.001 mm
- ☒ 0.001 cm

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Answered and Marked for Review (will
be considered for evaluation)**PHYSICS SINGLE CORRECT**

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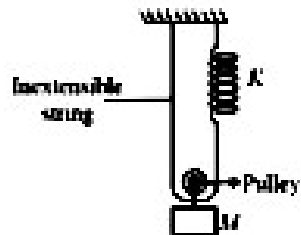
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The time period of mass M when displaced from its equilibrium position and then released for the system as shown in figure is



☐ $2\pi\sqrt{\frac{M}{k}}$

☐ $2\pi\sqrt{\frac{M}{2k}}$

☒ $2\pi\sqrt{\frac{M}{4k}}$

☐ $2\pi\sqrt{\frac{2M}{k}}$

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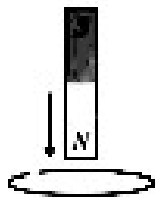
Question No. 15 #665df48dca324ed4a4c50f0be15

[Section Details](#) [Marking Scheme](#)

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A copper ring having a cut such as not to form a complete loop is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring, figure. The acceleration of the falling magnet is

☒ g

- ☐ Less than g
- ☐ More than g
- ☐ Zero

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PHYSICS SINGLE CORRECT

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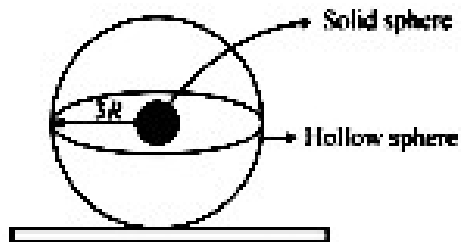
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This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A solid ball of mass m and radius R is released from the position shown in a large hollow fixed shell of same mass m and radius $3R$ as shown in the figure. The displacement of the centre of mass of the system from its initial position when the solid ball touches the lower surface of the hollow shell is (centres of both the spheres coincide initially):



- ☐ 0
- ☐ $3R$
- ☐ $2R$
- ☒ R

[Mark for Review & Next](#)[Clear Response](#)**PHYSICS SINGLE CORRECT**

Choose a question

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Choose a question

A plane electromagnetic wave travelling along the x -direction has a wavelength of 3 mm. The variation in the electric field occurs in the y -direction with an amplitude 66 Vm^{-1} . The equation for the electric and magnetic fields as a function of x and t are respectively

- ☐ $E_y = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$ and
 $B_z = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- ☐ $E_y = 11 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$ and
 $B_z = 11 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- ☐ $E_x = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$ and
 $B_y = 11 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c} \right)$
- ☒ $E_y = 66 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$ and
 $B_z = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c} \right)$

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Question No. 18 #665d746dca324ed4a4c501be18

[Section Details](#)[Marking Scheme](#)

Physics Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of these is correct match?

A.	Angular momentum	1.	$[M^{-1}L^2T^{-2}]$
B.	Torque	2.	$[M^1T^{-2}]$
C.	Gravitational constant	3.	$[M^1L^2T^{-2}]$
D.	Tension	4.	$[M^1L^2T^{-1}]$

- ☐ C - 2, D - 1
- ☒ A - 4, B - 3
- ☐ A - 3, C - 2
- ☐ B - 2, A - 1

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Answered and Marked for Review (will be considered for evaluation)

PHYSICS SINGLE CORRECT

Choose a question

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Question No. 19 #665df48dca324ed4a4c50fbc19

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

n identical resistance are taken in which $\frac{n}{2}$ resistors are joined in series in the left gap and the remaining $\frac{n}{2}$ resistances are joined in parallel in the right gap of a metre bridge. Balancing length in cm is

- ☒ $100 \cdot \frac{n^2}{n^2+4}$
- ☐ $100 \cdot \frac{n^2}{n^2+1}$
- ☐ $400 \cdot \frac{1}{n^2+4}$
- ☐ $400 \cdot \frac{1}{n^2+1}$

Mark for Review & Next

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PHYSICS SINGLE CORRECT

Choose a question

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Question No. 20 #665df48dca324ed4a4c50fbe20

[Section Details](#) < [Marking Scheme](#) >

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) When an ideal gas is compressed adiabatically, its temperature and the average kinetic energy of the gas molecules increase.

Reason (R) The kinetic energy increases because of collisions of molecules with moving parts of wall only.

- ☒ A and R are true and R is correct explanation of A.
- ☐ A and R are true but R is not correct explanation of A.
- ☐ A is true and R is false.
- ☐ A is false and R is true.

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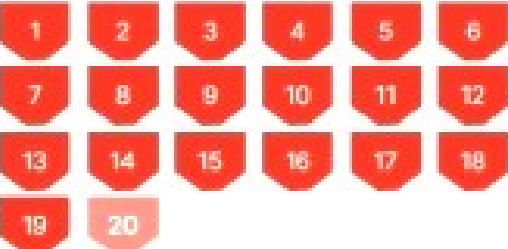
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PHYSICS SINGLE CORRECT

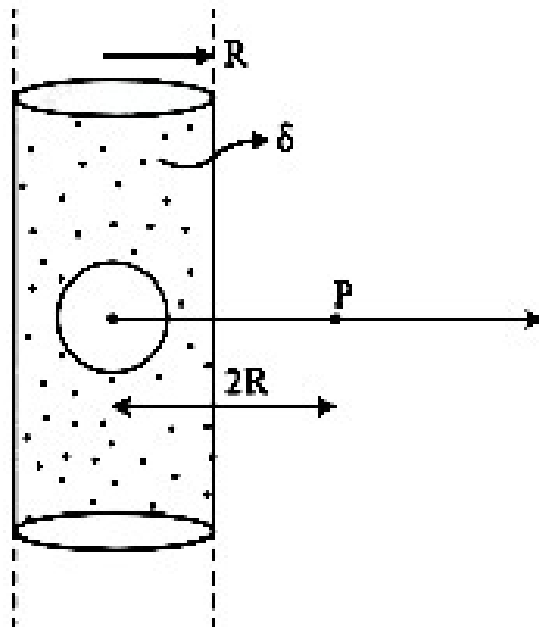
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This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

An infinitely long solid cylinder of radius R with uniform volume charge density ρ has a spherical cavity of radius $\frac{R}{2}$ with its centre on the axis of the cylinder as shown in the figure. The magnitude of the electric field at a point P which is at a distance $2R$ from the axis of the cylinder is given by $\frac{23\rho R}{6K\epsilon_0}$. What is the value of K ?



Ans: 16

PHYSICS NUMERICAL

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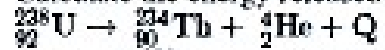
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Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Calculate the energy released in MeV in the following nuclear reaction :



[Mass of ${}_{92}^{238}\text{U} = 238.05079 \text{ u}$

Mass of ${}_{90}^{234}\text{Th} = 234.043630 \text{ u}$

Mass of ${}_2^4\text{He} = 4.002600 \text{ u}$

1 u = 931.5 MeV (mark answer to nearest integer in MeV)

PHYSICS NUMERICAL

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Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Three identical uniform thin metal rods form the three sides of an equilateral triangle. If the moment of inertia of the system of these three rods about an axis passing through the centroid of the triangle and perpendicular to the plane of the triangle is n times the moment of inertia of one rod separately about an axis passing through the centre of the rod and perpendicular to its length, the value of n is

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PHYSICS NUMERICAL

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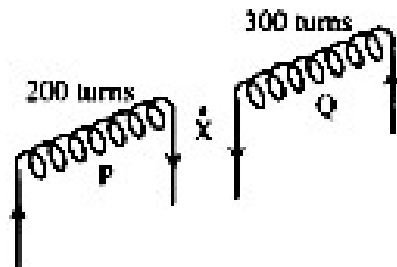
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Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Two solenoids P and Q of equal length but different number of turns, are arranged coaxially as shown in figure. P has 200 turns and Q has 300 turns. There is a current of 1 A in Q , the current in P in order that there is no resultant field at X midway between the coils is K , find $8 K$.



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Answered and Marked for Review (will be considered for evaluation)

PHYSICS NUMERICAL

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Question No. 5 #565df148dca324ed4a4c50fbc5

[Section Details](#) [Marking Scheme](#)

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Bulk modulus of water is $2 \times 10^9 \text{ N/m}^2$. The pressure required to increase the volume of water by 0.1% in N/m^2 is $2 \times 10^K \text{ N/m}^2$. Find K.

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be considered for evaluation)

PHYSICS NUMERICAL

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Question No. 1 #665df48dca324ed4a4c50fbc1

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

RNA and DNA are chiral molecules, their chirality is due to

- ☐ chiral bases
- ☐ chiral phosphate ester units
- ☒ D-sugar component
- ☐ L-sugar component

Mark for Review & Next

Clear Response

Save & Next

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19

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Answered and Marked for Review (will
be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Activate Windows
Go to Settings to activate Windows.

Question No. 2 #665df48dca324ed4a4c50fbe2

[Section Details](#) < [Marking Scheme](#) >

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

$\text{Mg}(\text{OH})_2$ is precipitated when NaOH is added to a solution of Mg^{2+} . If the final concentration of Mg^{2+} is 10^{-10} M , the concentration of $\text{OH}^- (\text{M})$ in the solution is

[Solubility product for $\text{Mg}(\text{OH})_2 = 5.6 \times 10^{-12}$]

☐ 0.056☐ 0.12☒ 0.24☐ 0.025

Mark for Review & Next

Clear Response

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18

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Activate Windows
Go to Settings to activate Windows.

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Match the following.

List I (Element)	List II (Property)
A. He	I. Can be used as inert atmosphere
B. Ar	II. Radioactive element
C. Xe	III. Form fluoride compounds
D. Rn	IV. Can be used as airship balloons

The correct match is

A B C D

☐ IV III II I

☒ IV I III II

Mark for Review & Next

Clear Response

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Activate Windows
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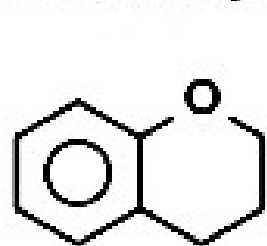
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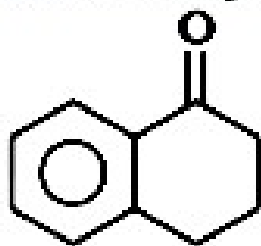
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

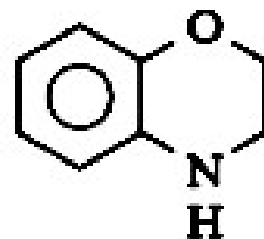
Rank the following compounds in decreasing order of reactivity in electrophilic aromatic substitution reaction.



(1)



(2)



(3)



(4)

☐ $3 > 1 > 2 > 4$ ☐ $4 > 3 > 2 > 1$ ☒ $3 > 1 > 4 > 2$ ☐ $1 > 3 > 4 > 2$

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Mark for Review & Next

Clear Response

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Activate Windows
Go to Settings to activate Windows.

Question No. 5 #665df48dca324ed4a4c50fbe5

[Section Details](#) < [Marking Scheme](#) >

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

In an atom the order of increasing energy of electrons with quantum numbers

(i) $n = 4, \ell = 1$

(ii) $n = 4, \ell = 0$

(iii) $n = 3, \ell = 2$ and (iv) $n = 3, \ell = 1$ is

☐ iii < i < iv < ii

☐ ii < iv < i < iii

☐ i < iii < ii < iv

☒ iv < ii < iii < i

[Mark for Review & Next](#)[Clear Response](#)[Save & Next](#)[Submit](#)

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Answered and Marked for Review (will
be considered for evaluation)**CHEMISTRY SINGLE CORRECT**

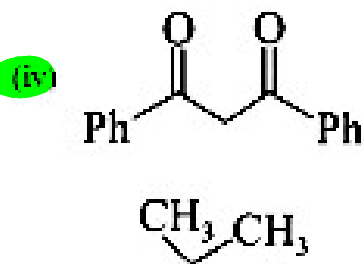
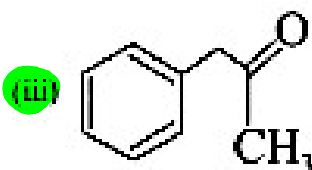
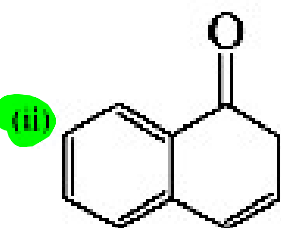
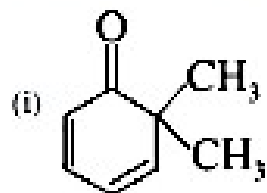
Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Activate Windows
Go to Settings to activate Windows.

Which of the following carbonyl compounds will exhibit enolization?



Mark for Review & Next

Clear Response

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Save & Next

Submit

Activate Windows
Go to Settings to activate Windows.

Question No. 7 #665df48dca324ed4a4c50fbe7

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

For BCl_3 , AlCl_3 and GaCl_3 the increasing order of ionic character is

- ☐ $\text{BCl}_3 < \text{AlCl}_3 < \text{GaCl}_3$
- ☐ $\text{GaCl}_3 < \text{AlCl}_3 < \text{BCl}_3$
- ☒ $\text{BCl}_3 < \text{GaCl}_3 < \text{AlCl}_3$
- ☐ $\text{AlCl}_3 < \text{BCl}_3 < \text{GaCl}_3$

Mark for Review & Next

Clear Response

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Activate Windows
Go to Settings to activate Windows.

Question No. 8 #665df48dca324ed4a4c50fbe8

[Section Details](#) < [Marking Scheme](#) >

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The boiling point of an azeotropic mixture of water and ethyl alcohol is less than that of theoretical value of water and alcohol mixture. Hence, the mixture shows

- ☐ that solution is highly saturated
- ☒ positive deviation from Raoult's law
- ☐ negative deviation from Raoult's law
- ☐ nothing can be said

Mark for Review & Next

Clear Response

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Answered and Marked for Review (will
be considered for evaluation)**CHEMISTRY SINGLE CORRECT**

Choose a question

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7	8	9	10	11	12
13	14	15	16	17	18
19	20				



Activate Windows
Go to Settings to activate Windows.

Question No. 9 #665df48dca324ed4a4c50fbe9

[Section Details](#) < [Marking Scheme](#) >

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which complex among the following has the highest value of spin only magnetic moment?



Mark for Review & Next

Clear Response

Save & Next

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11

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

Activate Windows
Go to Settings to activate Windows.

Question No. 10 #665df48dca324ed4a4c50fbe10

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Directions : Each of these questions contains an Assertion followed by reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

Assertion: *o*-Hydroxy benzene sulphonic acid on heating to 373 K changes to *p*-Hydroxybenzene sulphonic acid.

Reason: Sulphonation of phenol is a reversible process.

- ☐ If both assertion and reason are true and reason is the correct explanation of assertion.
- ☒ If both assertion and reason are true but reason is not the correct explanation of assertion.
- ☐ If Assertion is true but reason is false.
- ☐ If both assertion and reason are false.

Mark for Review & Next

Clear Response

Save & Next

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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7	8	9	10	11	12
13	14	15	16	17	18
19	20				

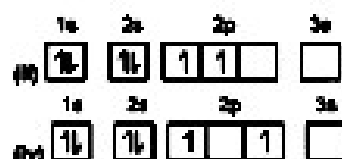
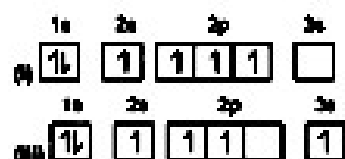
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Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Consider the following configurations:



Which of the above configuration may represent the excited state of C-atom.

- ☒ (i), (iii) & (iv)
- ☐ (i) only
- ☐ (i) & (iii)
- ☐ (iii) only

Mark for Review & Next

Clear Response

Save & Next

Submit



Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
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Activate Windows
Go to Settings to activate Windows.

Question No. 12 #665df48dca324ed4a4c50fbc12

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) Boron has a smaller first ionisation enthalpy than beryllium.

Reason (R) The penetration of a 2s-electron to the nucleus is more than the 2p-electron hence, 2p-electron is more shielded by the inner core of electrons than 2 s-electrons.

- ☐ Assertion and Reason both are correct statements but Reason is not correct explanation for Assertion.
- ☐ Assertion is correct statement but Reason is wrong statement.
- ☒ Assertion and Reason both are correct statements and Reason is correct explanation for Assertion.
- ☐ Assertion and Reason both are wrong statements.

Mark for Review & Next

Clear Response

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

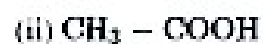
1	2	3	4	5	6
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19	20				

Activate Windows
Go to Settings to activate Windows.

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The order of strengths of the following carboxylic acids is



☒ (iii) > (iv) > (ii) > (i)

☐ (iv) > (ii) > (iii) > (i)

☐ (iii) > (i) > (iv) > (ii)

☐ (i) > (iv) > (ii) > (iii)

Mark for Review & Next

Clear Response

CHEMISTRY SINGLE CORRECT

Choose a question

1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

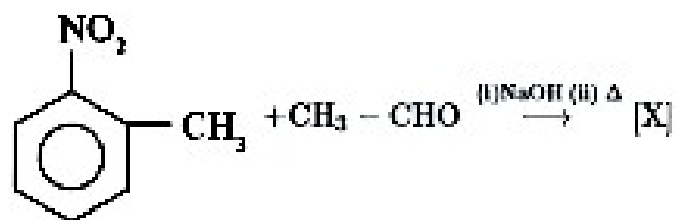


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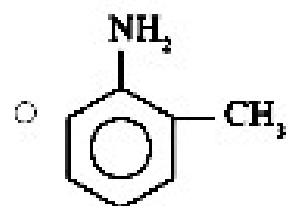
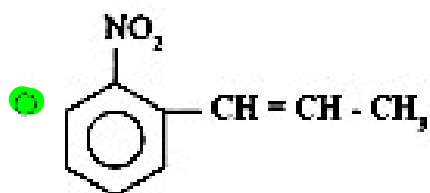
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In the given reaction



Then [X] will be



Mark for Review & Next

Clear Response



1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20				

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Save & Next

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Question No. 15 #665df48dca324ed4a4c50fbc15

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following is not correct about Grignard reagent?

- ☐ It is a nucleophile
- ☐ Forms new carbon-carbon bond
- ☐ Reacts with carbonyl compounds
- ☒ It is an organomanganese compound

[Mark for Review & Next](#)[Clear Response](#)[Save & Next](#)[Submit](#)

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Answered and Marked for Review (will
be considered for evaluation)**CHEMISTRY SINGLE CORRECT**

Choose a question



Activate Windows
Go to Settings to activate Windows.

Question No. 16 #665df48dca324ed4a4c50fbc16

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The strongest Lewis acid is -

- ☐ BF_3
- ☐ BCl_3
- ☐ BBr_3
- ☒ BI_3

Mark for Review & Next

Clear Response

Save & Next

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0 Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Question No. 17 #665df48dca324ed4a4c50fbc17

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A compound *A* contains *C* and *H* only and has molecular mass 72. It's photochlorination gives a mixture containing only one monochloro and two dichloro hydrocarbons. The IUPAC name is

☒ 2, 2-dimethylpropane

☐ n-pentane

☐ 2-methyl butane

☐ cyclopentane

[Mark for Review & Next](#)[Clear Response](#)[Save & Next](#)[Submit](#)

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Answered and Marked for Review (will
be considered for evaluation)**CHEMISTRY SINGLE CORRECT**

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 18 #665df48dca324ed4a4c50fbc18

[Section Details](#) [Marking Scheme](#)

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following statement(s) is (are) correct when a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ? Mark incorrect option

- ☐ A deep red vapour is evolved
- ☐ The vapour when passed through NaOH solution, gives a yellow solution
- ☒ Chlorine gas is also evolved
- ☐ Chromyl chloride is formed

Mark for Review & Next

Clear Response

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 19 #665df48dca324ed4a4c50fbc19

[Section Details](#) < [Marking Scheme](#) >

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

2 -Bromobutane (A) is treated with NaI in the presence of dry acetone to give compound 'B'. The compound 'B' is boiled with moist silver oxide to give compound 'C'. Identify the compounds 'B' and 'C'. How many optical isomers are possible for compound 'C'?

- ☒ 2 -Iodobutane, Butan-2 -ol, 2 isomers
- ☐ 3 -Iodopentane, Pentan-3 -ol, 4 isomers
- ☐ 2 -Iodobutane, Butan-2 -ol, 4 isomers
- ☐ 3 -Iodopentane, Pentan-3 -ol, 2 isomers

Mark for Review & Next

Clear Response

Save & Next

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 20 #665df48dca324ed4a4c50fbc20

Section Details

Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Choose the correct option regarding the following.

Assertion : Actinides exhibit a larger number of oxidation states than lanthanides.

Reason : There is a large energy gap between $5f$, $6d$ and $7s$ sub-shells.

☐ Assertion and Reason are correct, and Reason is the correct explanation for the Assertion.

☐ Assertion and Reason are correct, and Reason is not the correct explanation for the Assertion.

☒ Assertion is correct, Reason is wrong.

☐ Assertion is incorrect, Reason is correct.

Mark for Review & Next

Clear Response

Save & Next

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY SINGLE CORRECT

Choose a question

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Question No. 1 #665df48dca324ed4a4c50fbc1

[Section Details](#) [Marking Scheme](#)

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If 5.79 A current for 20000 sec. is passed in electrolysis of 0.3 molar, 1 litre CuSO_4 (aq) using Pt electrode then ratio of volume of gas liberated at cathode and anode at STP is $x : 1$. The value of x is

1

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Mark for Review & Next

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 2 #665df48dca324ed4a4c50fbe2

[Section Details](#) [Marking Scheme](#)Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

The concentration of R in the reaction $R \rightarrow P$ was measured as a function of time and the following data is obtained:

$[R]$ (molar)	1.0	0.75	0.40	0.10
t (min).	0.0	0.05	0.12	0.18

The order of the reaction is

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4	5	6
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Mark for Review & Next

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 3 #665df48dca324ed4a4c50fbc3

[Section Details](#) [Marking Scheme](#)

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

During the Kjeldahl method, 2.8 g of an organic compound was digested and the evolved ammonia was absorbed by 60 mL of $\frac{M}{20}$ H_2SO_4 . What volume of $\frac{M}{20}$ NaOH solution was required for complete neutralisation of the unreacted acid if the percentage of nitrogen in the compound was 1% ? (mark answer in mL)

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Mark for Review & Next

Clear Response

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 4 #665df48dca324ed4a4c50fbe4

[Section Details](#) [Marking Scheme](#)

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A metal (M) forms two oxides. The ratio $M : O$ (by weight) in the two oxides are $25 : 4$ and $25 : 6$. The minimum value of atomic mass of M is

Backspace

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Mark for Review & Next

Clear Response

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Answered and Marked for Review (will be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 5 #665df48dca324ed4a4c50fbe5

[Section Details](#) [Marking Scheme](#)

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Two mole of an ideal gas is isothermally compressed in two stages starting from (1 atm, 20 lit) to 10 lit. at external pressure P_1 then upto 2 lit at external pressure P_2 . Find work done (lit-atm) in the process.

Backspace

7 8 9

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Mark for Review & Next

Clear Response

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Marked for
ReviewAnswered and Marked for Review (will
be considered for evaluation)

CHEMISTRY NUMERICAL

Choose a question

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Activate Windows
Go to Settings to activate Windows.

Question No. 1 #665df48dca324ed4a4c50fbc1

Section Details > Marking Scheme >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If L and M are respectively the coefficient of x^{-7} in $\left(ax + \frac{b}{x^2}\right)^{11}$ and the coefficient of x^7 in $\left(bx^2 + \frac{a}{x}\right)^{11}$ then $L + M =$

- ☐ $\frac{1}{b} \left[\text{coefficient of } x^{-6} \text{ in } \left(ax + \frac{b}{x^2}\right)^{12} \right]$
- ☒ $\frac{1}{a} \left[\text{coefficient of } x^6 \text{ in } \left(ax^2 + \frac{b}{x}\right)^{12} \right]$
- ☐ $a \left[\text{coefficient of } x^{-10} \text{ in } \left(ax + \frac{b}{x^2}\right)^{11} \right]$
- ☐ $b \left[\text{coefficient of } x^4 \text{ in } \left(ax^2 + \frac{b}{x}\right)^{11} \right]$

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MATHEMATICS SINGLE CORRECT

Choose a question

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Question No. 2 #665df48dca324ed4a4c50fbe2

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If the foot of the perpendicular from the point $A(p+1, -1, 11)$ on the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is $B(q, 5, 7)$ then the value of $(p - q)$ is

- ☐ 1.5
- ☐ 1
- ☐ 0.5
- ☒ 0

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Answered and Marked for Review (will
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MATHEMATICS SINGLE
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Question No. 3 #665df48dca324ed4a4c50fbe3

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The particular solution of the different equation

$$\frac{dx}{dy} = \frac{\sin y(1-y \cot y)}{x \log(x^2 e)}, y(1) = 0$$

- ☒ $y \sin y = x^2 \log x$
- ☐ $y^2 \sin y = \log x$
- ☐ $y = \left(\frac{e^2}{\sin x} \right) (x - 1)$
- ☐ $y = e^2 \sec x$

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Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE
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Question No. 4 #665df48dca324ed4a4c50fba4

Section Details > Marking Scheme >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $J = \int_0^{e-1} \frac{1}{x+1} \exp \cdot \left(\frac{x^2+2x-1}{2} \right) dx$ and $K = \int_1^e x \ln x \exp \cdot \left(\frac{x^2-2}{2} \right) dx$. The value of $(J + K)$ is equal to (Here exp. (x) means e power x)

- ☐ $(\sqrt{e})^{e^2+1}$
- ☐ $(\sqrt{e})^{e^2-1}$
- ☐ 0

☒ $(\sqrt{e})^{e^2-2}$

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Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE
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Question No. 5 #665df48dca324ed4a4c50fbe5

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If sum of coefficients in the expression of $(\alpha x^2 - 2x + 1)^{35}$ is equal to sum of coefficients in the expansion of $(x - \alpha y)^{35}$, then α is equal to

- ☐ 0
- ☒ 1
- ☐ any real number.
- ☐ 2

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Answered and Marked for Review (will
be considered for evaluation)

**MATHEMATICS SINGLE
CORRECT**

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Question No. 6 #665df48dca324ed4a4c50fbe6

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The number of common terms between the sequences given by $1, 4, 7, 10, \dots, 298$ and $2, 4, 6, 8, \dots, 300$ is

- ☐ 49
- ☒ 50
- ☐ 48
- ☐ 100

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Answered and Marked for Review (will
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**MATHEMATICS SINGLE
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Question No. 7 #665df48dca324ed4a4c50fbe7

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The axis of a parabola lie along the line $y = x$ and the distance of its vertex from origin is $\sqrt{2}$ and that of focus is $2\sqrt{2}$. If both focus and vertex lie in the first quadrant, then the equation of the parabola will be

- ☐ $(x + y)^2 = (x - y - 2)$
- ☐ $(x - y)^2 = (x + y - 2)$
- ☒ $(x - y)^2 = 4(x + y - 2)$
- ☐ $(x - y)^2 = 8(x + y - 2)$

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Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE
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Question No. 8 #665df48dca324ed4a4c50fbe8

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The reciprocal of the distance between two points, one on each of the lines $\frac{x-2}{3} = \frac{y-4}{2} = \frac{z-6}{5}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$,

Then which of the following is FALSE.

- ☐ can be 8
- ☐ can be less than $\sqrt{75}$
- ☐ cannot be greater than 78
- ☒ can be $\sqrt{80}$

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Answered and Marked for Review (will
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MATHEMATICS SINGLE
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Question No. 9 #665df48dca324ed4a4c50fbe9

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be an even function. $I_1 = \int_0^{\frac{\pi}{2}} f(\cos 2x) \cos x dx$, $I_2 = \int_0^{\frac{\pi}{4}} f(\sin 2x) \cos x dx$. Then $\frac{I_1}{I_2} =$

- ☐ 1
- ☐ $\frac{1}{2}$
- ☐ $\frac{1}{\sqrt{2}}$
- ☒ $\sqrt{2}$

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Answered and Marked for Review (will be considered for evaluation)

**MATHEMATICS SINGLE
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Choose a question

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Question No. 10 #665df48dca324ed4a4c50fba10

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A relation R is defined as $(x, y) \in R \Rightarrow x^y = y^x$ for $x, y \in I - \{0\}$, where I is the set of all integers. Then the relation R is:

- ☐ reflexive but not symmetric
- ☐ symmetric but not reflexive
- ☒ reflexive and symmetric both
- ☐ equivalence relation

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MATHEMATICS SINGLE
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Choose a question

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Question No. 11 #665df48dca324ed4a4c50fbc11

[Section Details](#) > [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{z}{z}$, then the locus traced by ' ω ' in the complex plane is

- ☒ $x - y - 1 = 0$
- ☐ $x + y - 1 = 0$
- ☐ $x - y + 1 = 0$
- ☐ $x + y + 1 = 0$

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9 Not Visited 0 Marked for Review

0 Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE
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Choose a question

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Question No. 12 #665df48dca324ed4a4c50fbc12

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The distinct points $A(0, 0)$, $B(0, 1)$, $C(1, 0)$ and $D(2a, 3a)$ are concyclic, then

- ☒ 'a' can attain only rational values
- ☐ 'a' is irrational
- ☐ Cannot be concyclic for any 'a'
- ☐ None of the above

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Answered and Marked for Review (will be considered for evaluation)

MATHEMATICS SINGLE
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Question No. 13 #665df48dca324ed4a4c50fbc13

Section Details > Marking Scheme >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be defined by $f(x) = \begin{cases} \frac{3x-[x]}{x}, & x < 1 \\ \frac{3-x^2+x}{|x|}, & x \geq 1 \end{cases}$

(where $[.]$ is G.I.F) Then which is incorrect

- ☐ $f(x)$ is continuous at $x = 1$
☐ $f(x)$ is not differentiable at $x = 1$
☐ $f'\left(\frac{3}{2}\right) = -2$
☒ $f'(x)$ is continuous at $x = 1$

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MATHEMATICS SINGLE
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Question No. 14 #665df48dca324ed4a4c50fbc14

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The mean and standard deviation of 10 observations $x_1, x_2, x_3, \dots, x_{10}$ are $\bar{x}$ and σ respectively. Let 10 is added to $x_1, x_2, \dots, x_9$ and 90 is subtracted from x_{10} . If still, the standard deviation is the same, then $x_{10} - \bar{x}$ is equal to

- ☐ 35
- ☒ 45
- ☐ 55
- ☐ 50

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MATHEMATICS SINGLE
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Question No. 15 #665df48dca324ed4a4c50fbc15

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The longest distance of the point $(a, 0)$ from the curve $2x^2 + y^2 = 2x$ is

- ☐ $1 + a$
- ☐ $|1 - a|$
- ☒ $\sqrt{1 - 2a + 2a^2}$
- ☐ $\sqrt{1 - 2a + 3a^2}$

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Answered and Marked for Review (will
be considered for evaluation)

MATHEMATICS SINGLE
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Question No. 16 #665df48dca324ed4a4c50fbc16

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = \cos^2 x$; $g(x) = \int_0^{1/x} e^t dt$. The value of $\lim_{x \rightarrow 0} (f(x))^{g(x)}$ is

- ☐ 1
- ☐ e
- ☐ e^{-1}
- ☒ 0

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MATHEMATICS SINGLE
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Question No. 17 #665df48dca324ed4a4c50fbc17

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

- ☐ 7
- ☒ -7
- ☐ -5
- ☐ 5

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MATHEMATICS SINGLE
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Question No. 18 #665df48dca324ed4a4c50fbc18

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, $a, b \in N$. Then :

- ☐ there exist more than one but finite number of B 's such that $AB = BA$
- ☐ there exists exactly one B such that $AB = BA$
- ☒ there exists infinitely many B 's such that $AB = BA$
- ☐ there cannot exist any B such that $AB = BA$

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MATHEMATICS SINGLE
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Question No. 19 #665df48dca324ed4a4c50fbc19

[Section Details](#) < [Marking Scheme](#) >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

$$f: R \rightarrow R \text{ is defined as } f(x) = \begin{cases} x^2 + 2mx - 1 & \text{for } x \leq 0 \\ mx - 1 & \text{for } x > 0 \end{cases}$$

If $f(x)$ is one-one then m must lie in the interval

- ☒ $(-\infty, 0)$
- ☐ $(-\infty, 0]$
- ☐ $(0, \infty)$
- ☐ $[0, \infty)$

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MATHEMATICS SINGLE
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Question No. 20 #665df48dca324ed4a4c50fbe20

[Section Details](#) [Marking Scheme](#)

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $\triangle ABC$ is an isosceles triangle with $AB = AC$. If $B = (0, a)$, $C = (2a, 0)$ and the equation of AB is $3x - 4y + 4a = 0$, then the equation of side AC is

- ☐ $y = 8x - 16a$
- ☐ $3y = 4x - 8a$
- ☒ $x = 2a$
- ☐ $y + 8x = 16a$

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MATHEMATICS SINGLE
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Question No. 1 #665df48dca324ed4a4c50fbc1

[Section Details](#) < [Marking Scheme](#) >**Mathematics Numerical (Maximum Marks: 20)**
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If the solution for the differential equation $y^2 dx + (x^2 - xy - y^2) dy = 0$ at $(2, 1)$ is $x + y = k(xy^2 - y^3)$, then $k =$

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ReviewAnswered and Marked for Review (will
be considered for evaluation)**MATHEMATICS NUMERICAL**

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Question No. 2 #665df48dca324ed4a4c50fbe2

[Section Details](#) < [Marking Scheme](#) >

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

$OABC$ is a unit square where O is the origin and $B = (1, 1)$. The curves $y^2 = x$ and $x^2 = y$ divide the area of the square into three parts $OABO$, OBO and $OBCO$. If a_1, a_2, a_3 are the areas (in sq units) of these parts respectively, then $a_1 + 2a_2 + 3a_3 =$

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MATHEMATICS NUMERICAL

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Question No. 3 #665df48dca324ed4a4c50fbe3

[Section Details](#) < [Marking Scheme](#) >**Mathematics Numerical (Maximum Marks: 20)**
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A biased coin that has a probability of getting heads as p ($0 < p < 1$) is tossed until a head appears for the first time. If the probability that the number of tosses required is even, is $\frac{2}{3}$, then $9p$ is equal to

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MATHEMATICS NUMERICAL

Choose a question

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Question No. 4 #665df48dca324ed4a4c50fbc4

Section Details  Marking Scheme 

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If $x = \operatorname{cosec} \theta - \sin \theta$, $y = \operatorname{cosec}^{2022} \theta - \sin^{2022} \theta$ and $\left(\frac{dy}{dx}\right)^2 = \frac{k(g^2+4)}{g(x)}$ where $k \in \mathbb{R}$, then $10 + k - g(2022) =$

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MATHEMATICS NUMERICAL

Choose a question

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Question No. 5 #665df48dca324ed4a4c50fbe5

[Section Details](#) < [Marking Scheme](#) >

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Number of skew-symmetric matrices of order 3 whose elements are $0, 0, 0, 1, -1, 2, -2, 3, -3$ is

Backspace

7 8 9

4 5 6

1 2 3

0 . -

Left Right

Clear All



Mark for Review & Next

Clear Response

Save & Next

Submit

0

Not Visited

0

Marked for
ReviewAnswered and Marked for Review (will
be considered for evaluation)

MATHEMATICS NUMERICAL

Choose a question

1

2

3

4

5

Activate Windows
Go to Settings to activate Windows.